ERMITTLUNGSAKTE · CASE FILE REFERENCE · EXCEL FOR THE WEB

TATORT: EXCEL

A Forensic Guide to Data Cleaning. Keep this next to you during the investigation, and afterwards.

THE INVESTIGATION

Step	Action	Why it matters
0 · Secure the Scene	Duplicate the raw sheet and work only on the copy. AutoSave is always on! Start a case log.	The original is your evidence. If anything goes wrong, you can always go back.
1 · Survey the Scene	Scroll everything. Freeze panes, turn on filters, open every drop-down. Note clues.	You can't fix what you haven't noticed. Look before you touch.
2 · Open the Case File	Write the order of operations. Log every change: what, why, how verified.	A plan makes the work repeatable, for colleagues and for future you.
3 · Clear the Scene	Delete junk rows above the header, irrelevant columns and exact duplicates.	Less to clean, fewer distractions.
4 · Process the Evidence	Fix spaces & hidden characters, data types, dates, case, missing-value placeholders.	Hidden problems break sorting, lookups, math and de-duplication.
5 · Identify the Suspects	Split combined columns, unmask aliases with a reference table, add validation.	One fact per column; one spelling per category.
6 · Cross-Examine	Remove duplicates again. Pivot tables, sorting and conditional formatting.	Confirms the cleaned data tells a consistent story.
7 · Close the Case	Save a clean copy, download a CSV, update the data dictionary, keep the case log.	Clean data is only useful if others can find, trust and reuse it.

WHERE TO FIND IT (EXCEL FOR THE WEB)

Tool	Location
Open a shared file	Open the link → File → Create a Copy → Create a copy online (your copy lives in your OneDrive)
Open your own file	office.com → Excel → Upload a file
Copy a sheet	Right-click sheet tab → Duplicate
Freeze header row	View → Freeze Panes → Freeze Top Row
Filter	Data → Filter (or Home → Sort & Filter)
Make a Table	Click in the data → Insert → Table. New formula columns fill down by themselves
Find blanks	Filter a column → tick only (Blanks); count with `=COUNTBLANK(range)`
Find & Replace	Home → Find & Select → Replace (Ctrl/⌘ + H) → Search options
Remove Duplicates	Data → Remove Duplicates
Highlight duplicates	Home → Conditional Formatting → Highlight Cells Rules → Duplicate Values
Split a column	`TEXTBEFORE` / `TEXTAFTER` / `TEXTSPLIT`, or search for *Split Text to Columns*
Paste as values	Copy, then **Ctrl/⌘ + Shift + V** (the Paste ▾ menu usually can't reach the clipboard in the browser)
Numbers stored as text	Helper column with `=VALUE()`, then Paste Values
Restrict entries	Data → Data Validation
Spot-check	Insert → PivotTable; Home → Conditional Formatting
Save a copy	File → Create a Copy → Create a copy online
Download a CSV	File → Export → Download as CSV UTF-8
Roll back a disaster	File → Version History
Can't find a button?	Type its name in the search box at the top of Excel

HANDY FUNCTIONS

Function	What it does	Example from the Tatort case
`TRIM`	Removes leading, trailing and double spaces	`=TRIM(B2)`
`CLEAN`	Removes non-printing characters (line breaks, …)	`=TRIM(CLEAN(B2))`
`SUBSTITUTE`	Replaces specific text inside a cell	`=SUBSTITUTE(G2, CHAR(10), ", ")`
`UNICHAR`	Any character by its Unicode number	`=TRIM(SUBSTITUTE(B2, UNICHAR(160), " "))`
`LEN`	Counts characters, including invisible ones	`=LEN(B2)`
`ISNUMBER` / `ISTEXT`	Reveals a cell's real data type	`=ISNUMBER(A2)`
`LET`	Gives a value a short name inside a formula	`=LET(d, [@Erstausstrahlung], IF(ISNUMBER(d), d, …))`
`VALUE`	Turns number-looking text into a number	`=VALUE(SUBSTITUTE(A2, "#", ""))`
`DATE`	Builds a real date from year, month, day	`=DATE(RIGHT(D2,4), MID(D2,4,2), LEFT(D2,2))`
`TEXT`	Formats a number/date as text	`=TEXT(D2, "yyyy-mm-dd")`
`UPPER` / `LOWER` / `PROPER`	Changes capitalisation	`=PROPER(B2)` (check the result!)
`EXACT`	Case-sensitive comparison	`=EXACT(B2, UPPER(B2))` flags ALL CAPS
`LEFT` / `RIGHT` / `MID`	Pulls characters from a position	`=LEFT(E2, FIND("(", E2) - 1)`
`FIND` / `SEARCH`	Position of text (case-sensitive / not)	`=FIND("(", E2)`
`TEXTBEFORE` / `TEXTAFTER`	Text before/after a delimiter	`=TEXTBEFORE(C2, "/", , , , C2)`
`TEXTSPLIT`	Splits into several cells	`=TEXTSPLIT(C2, "/")`
`CONCAT` / `&` / `TEXTJOIN`	Combines text	`=A2 & " – " & B2`
`IFERROR`	Fallback when a formula errors	`=IFERROR(LEFT(E2, FIND("(",E2)-1), E2)`
`XLOOKUP` / `VLOOKUP`	Looks up a value in a reference table	`=XLOOKUP(C2, Known_Aliases!A:A, Known_Aliases!B:B, "CHECK")`
`COUNTIF`	Counts matches: finds duplicates	`=COUNTIF(A:A, A2) > 1`
`ROUND` / `ROUNDUP` / `ROUNDDOWN`	Changes the stored value, not just the display	`=ROUND(H2, 0)`

KNOWN TRICKS OF THE TRADE

CAREFUL

TRIM and CLEAN miss the non-breaking space (character 160). Use `SUBSTITUTE(…, UNICHAR(160), " ")` first. `UNICHAR` means the same thing everywhere.

CAREFUL

Find & Replace replaces text inside cells (`-` in *Schulze-Rohr*!) and treats `?` and `*` as wildcards. For placeholders, use `=IF(OR(G2="n/a", G2="-"), "", G2)` instead.

CAREFUL

`DATEVALUE` follows your computer's regional settings. `DATE(year, month, day)` never guesses, so build dates explicitly.

CAREFUL

`PROPER` doesn't speak German (*Taxi nach Leipzig*) and breaks genuine acronyms (*KI*). Flag, then review.

CLUE

Same value ≠ duplicate. Three different episodes are called *Die Abrechnung*. Decide which columns define "the same record".

CLUE

Helper-column routine: add a column to the Table → formula (it fills down) → check (`LEN` , filters, counts) → keep it beside the original. At the very end, in a copy: Paste Values over the original, delete helpers, log it.

CLUE

In a Table, `[@Titel]` = this row's `Titel` ; `[Titel]` = the whole column. Click a cell while writing a formula and Excel writes the reference for you.

CLUE

Clean, then de-duplicate again. Near-duplicates only match once spaces and spellings are fixed.